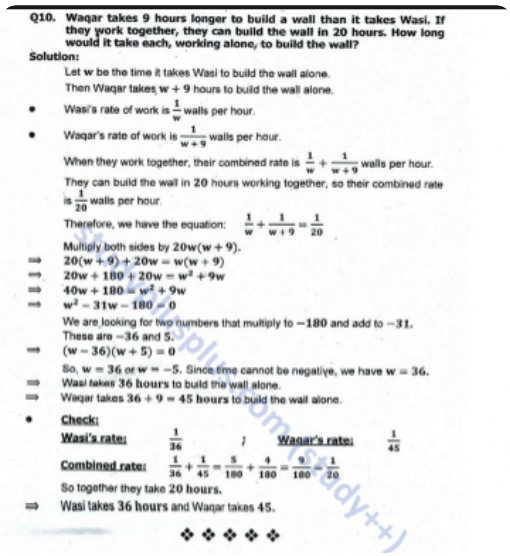


Q10. **Waqar takes 9 hours longer to build a wall than it takes Wasi. If they work together, they can build the wall in 20 hours. How long would it take each, working alone, to build the wall?**

Solution:

Let w be the time it takes Wasi to build the wall alone.

Then Waqar takes $w + 9$ hours to build the wall alone.

- Wasi's rate of work is $\frac{1}{w}$ walls per hour.
- Waqar's rate of work is $\frac{1}{w+9}$ walls per hour.

When they work together, their combined rate is $\frac{1}{w} + \frac{1}{w+9}$ walls per hour.

They can build the wall in 20 hours working together, so their combined rate is $\frac{1}{20}$ walls per hour.

Therefore, we have the equation: $\frac{1}{w} + \frac{1}{w+9} = \frac{1}{20}$

Multiply both sides by $20w(w+9)$.

$\Rightarrow 20(w+9) + 20w = w(w+9)$

$\Rightarrow 20w + 180 + 20w = w^2 + 9w$

$\Rightarrow 40w + 180 = w^2 + 9w$

$\Rightarrow w^2 - 31w - 180 = 0$

We are looking for two numbers that multiply to -180 and add to -31.

These are -36 and 5.

$\Rightarrow (w - 36)(w + 5) = 0$

So, $w = 36$ or $w = -5$. Since time cannot be negative, we have $w = 36$.

$\Rightarrow$ Wasi takes 36 hours to build the wall alone.

$\Rightarrow$ Waqar takes $36 + 9 = 45$ hours to build the wall alone.

- **Check:**

Wasi's rate: $\frac{1}{36}$; **Waqar's rate:** $\frac{1}{45}$

Combined rate: $\frac{1}{36} + \frac{1}{45} = \frac{5}{180} + \frac{4}{180} = \frac{9}{180} = \frac{1}{20}$

So together they take 20 hours.

$\Rightarrow$ Wasi takes 36 hours and Waqar takes 45.

❖ ❖ ❖ ❖ ❖

explain why we take reciprocal of the values

Great question 👍 Umair, yeh concept samajh lo to “work & time” ke sare sawal easy ho jate hain.

🔁 Reciprocal kyun lete hain?

Jab question **time** deta hai (jaise 36 hours, 45 hours), lekin hume **rate of work** chahiye hota hai, to hum us ka reciprocal lete hain.

📌 Basic Concept

Agar koi banda:

- 36 ghanton me **1 poora kaam** karta hai
  👉 to 1 ghante me kitna karega?

Obviously:

$$\frac{1 \text{ work}}{36 \text{ hours}} = \frac{1}{36}$$

Is liye:

Time ka reciprocal = Rate of work

$$\text{Rate} = \frac{1}{\text{Time}}$$

🔎 Simple Example

Agar Waqas 45 ghante me 1 wall banata hai:

Matlab:

- 45 hours → 1 wall
- 1 hour → ? wall

To hum divide karte hain:

$$1 \div 45 = \frac{1}{45}$$

Is liye uska rate hai:

$$\frac{1}{45} \text{ wall per hour}$$

🤝 Jab Dono Saath Kaam Karen

Agar:

- Wasi ka rate = 1/36
- Waqas ka rate = 1/45

To dono mil kar:

$$\frac{1}{36} + \frac{1}{45}$$

Rate add hota hai, **time nahi**.

💡 Intuitive Soch

Socho:

Agar koi banda 10 ghante me kaam karta hai
To obviously wo 1 ghante me 1/10 kaam karega.

Agar 5 ghante me karta hai
To 1 ghante me 1/5 karega (zyada fast)

Is liye:

👉 Jitna kam time, utna zyada rate

👉 Is relation ko mathematically reciprocal dikhata hai

🔥 Short Rule Yaad Rakhne Ka

Work problems me:

- **Time diya ho → reciprocal le ke rate banao**
- **Saath kaam karein → rates add karo**
- **Phir final time nikalne ke liye dobara reciprocal lo**

Agar chaho to main tumhe ek visual diagram ke sath aur clear kar dun 👍

why we take reciprocal if we have time to get rate and why we take again reciprocasl to get final time means once we take then why again also

if we have a price of 5 kg apples is 700 and want to get know the price of 1 kg we dont divide 1 by 5 even we 5 kg = 700 so divide both side by 5 know the price of 1 kg

Excellent question 👌 this means you're thinking logically, not just memorizing.

Let’s clear this properly.

🔷 1️⃣ Why do we take reciprocal in work problems?

Because **work problems are about “1 whole work”**.

When a question says:

> A man finishes a job in 5 hours

It means:

5 hours → 1 complete work

But we want **work per 1 hour**.

So we convert:

$$1 \text{ work} \div 5 \text{ hours} = \frac{1}{5}$$

That's not a random reciprocal.
It is simply:

$$\text{Rate} = \frac{\text{Work}}{\text{Time}}$$

Since work = 1, it becomes:

$$\frac{1}{5}$$

So reciprocal comes naturally from the formula.

🔷 2 Why take reciprocal AGAIN at the end?

When two people work together, we ADD their rates.

Example:

A = 1/5
B = 1/10

Together:

$$\frac{1}{5} + \frac{1}{10} = \frac{3}{10}$$

This means:

They do **3/10 work in 1 hour**

But we want:

Time to complete 1 full work.

So we solve:

$$\text{Time} = \frac{\text{Work}}{\text{Rate}}$$

$$= \frac{1}{3/10}$$

And dividing by a fraction means multiplying by its reciprocal:

$$= \frac{10}{3}$$

So we are not randomly taking reciprocal twice.

We are doing:

Rate = Work ÷ Time
Time = Work ÷ Rate

That's why reciprocal appears again.

🔷 3️⃣ Why apples example is different?

Excellent comparison 👏

You said:

5 kg = 700
So 1 kg = 700 ÷ 5

Correct.

But notice something:

Here you are dividing total price by total quantity.

You are using:

$$\text{Unit price} = \frac{\text{Total price}}{\text{Total kg}}$$

This is the SAME logic.

If you rewrite apples problem like work problem:

5 kg → 700
1 kg → ?

You divide by 5.

In work problems:

5 hours → 1 work
1 hour → ?

You divide by 5.

Same idea.

The only difference:

In apples, we divide 700 ÷ 5
In work, we divide 1 ÷ 5

That's why it looks like reciprocal — because numerator is 1.

🔥 Big Insight

Reciprocal happens naturally when:

The total amount = 1

If apples question was:

5 kg → 1 dollar

Then price of 1 kg = 1/5 dollar

Same reciprocal would appear.

🎯 Final Understanding

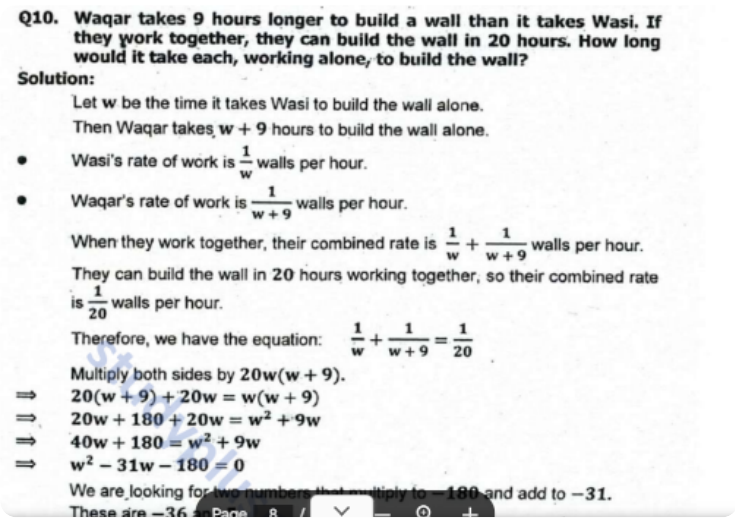
We don't "take reciprocal twice".

We:

1. Convert time → rate
2. Add rates
3. Convert rate → time

It's just dividing by something that is less than 1, which creates reciprocal automatically.

If you want, I can show this using a speed example (car km/h) which makes it even clearer.



Q10. Waqar takes 9 hours longer to build a wall than it takes Wasi. If they work together, they can build the wall in 20 hours. How long would it take each, working alone, to build the wall?

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$\Rightarrow \quad w^2 - 31w - 180 = 0$

We are looking for two numbers that multiply to -180 and add to -31.

These are -36 and

here if we know waqar and wasi total time then why we dont just simple add up and get each time or why we need the per hour time

Very strong question 👏 this is the exact point where most students get confused.

You're asking:

> If together they take 20 hours, why can't we just add their times and find each time?

Let's break this carefully.

🔷 1️⃣ Time does NOT add — Work Rate adds

When two people work together:

They are working at the SAME time.

So what increases?

👉 The **amount of work done per hour**

NOT the time.

🔷 2️⃣ Simple Real-Life Example

Suppose:

Wasi alone → 10 hours
Waqar alone → 10 hours

If we “add times”:

10 + 10 = 20 hours ❌

But is that correct?

No.

If both work together, they finish faster, not slower.

Actually:

Each does 1/10 work per hour.

Together:

1/10 + 1/10 = 2/10 = 1/5 work per hour

So full work takes:

5 hours ✅

See?

Time didn’t add.
Speed (rate) added.

🔷 3️⃣ Why We Need Per Hour Time

Because:

Work problems follow this rule:

Rate adds, Time doesn’t

Mathematically:

$$\text{Rate} = \frac{1}{\text{Time}}$$

So to combine two workers:

Step 1 → Convert time to rate
Step 2 → Add rates
Step 3 → Convert back to time

🔷 4️⃣ Why Can't We Just Add Their Times?

Because their times are for completing the ENTIRE job alone.

When working together:

Each person is contributing only part of the work.

So 20 hours together does NOT mean:

Wasi did 20 hours of full work
Waqar did 20 hours of full work

They each did partial work.

🔷 5️⃣ Think of Speed Example (Best Way to Understand)

If:

Car A speed = 60 km/h
Car B speed = 40 km/h

Driving together pulling one load:

Total speed = 60 + 40 = 100 km/h

We add speeds, not travel times.

Work problems are exactly the same as speed problems.

Work rate = speed
Total work = distance

🔥 Final Clear Answer

We don’t add times because:

Time tells how long ONE person needs to finish full job alone.

But when two people work together:

They combine their speeds (work per hour).

That’s why we must convert to per hour (rate).

If you want, I can show you a small visual table proof that makes it mathematically impossible to add times.



> if i have 8 apple and 2 banas and if i want to full the target i need 80% more bananas then how we compute this that how many bananas we need step by step in simple and easy way

Chalo step-by-step, bilkul simple 👇

Given

- Apples = 8
- Bananas (abhi) = 2
- Target: **bananas 80% more** chahiye (matlab current bananas se 80% zyada)

Step 1: 80% ka matlab samjho

80% = 80/100 = 0.8

Step 2: Current bananas ka 80% nikaalo

Current bananas = 2

$$80\% \text{ of } 2 = 0.8 \times 2 = 1.6$$

Step 3: Ye extra bananas hain jo add karne hain

Extra bananas = **1.6**

Step 4: Total bananas needed

$$2 + 1.6 = 3.6$$

Step 5: Real life me banana half nahi hota (usually)

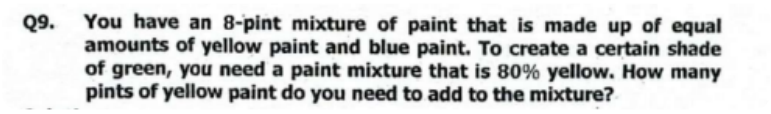
To aap **round up** karoge:

✅ **4 bananas**

Final Answer

Target complete karne ke liye 4 bananas chahiye (kyunke 3.6 possible nahi, so 4).

Agar aap chahein to main isi example ko “80% bananas compared to apples” wali condition me bhi solve karke dikha doon (kyunke wording kabhi kabhi us taraf bhi hoti hai).



simple and way tell me

Chalo bilkul simple tareeke se samajhte hain 👇

Given:

Total mixture = **8 pints**

Equal amounts of 3 colors (green, yellow, blue)

Step 1: Har color kitna hai?

3 colors equal hain.

$$8 \div 3 = 2.67 \text{ pints (approx)}$$

So:

Yellow ≈ 2.67 pints

Step 2: Ab hume 80% yellow chahiye

Abhi total mixture 8 pints hai.

Lekin hum yellow add karenge, to total badh jayega.

Maan lo hum **x pints yellow** add karte hain.

New yellow amount:

$$2.67 + x$$

New total mixture:

$$8 + x$$

Step 3: Condition lagao (80%)

Yellow should be 80% of total:

$$\frac{2.67 + x}{8 + x} = 0.8$$

Step 4: Solve simple way

Multiply both sides by (8 + x):

$$2.67 + x = 0.8(8 + x)$$

$$2.67 + x = 6.4 + 0.8x$$

Now rearrange:

$$x - 0.8x = 6.4 - 2.67$$

$$0.2x = 3.73$$

$$x = 18.67$$

Final Answer:

You must add **about 18.7 pints of yellow paint**.

Simple Understanding

Because starting yellow amount bohat kam tha (sirf 2.67),
aur aapko final mixture ka 80% yellow chahiye,
isliye bohat zyada yellow add karna padta hai.

Agar chaho to main isko aur bhi short trick se samjha doon jo exam me fast use hoti hai.

only 2 colors are given equal amount of yellow and blue and to need a green shade we need 80% yellow so let me know how many yellow pints add more

Chalo bilkul simple tareeke se 👇

Given:

Total mixture = **8 pints**
Sirf 2 colors hain: **Yellow aur Blue**
Equal amounts hain.

Step 1: Har color kitna hai?

$$8 \div 2 = 4$$

So:

- Yellow = 4 pints
- Blue = 4 pints

Step 2: Ab hume 80% yellow chahiye

Maan lo hum **x pints yellow** add karte hain.

New yellow amount:

$$4 + x$$

New total mixture:

$$8 + x$$

Condition:

$$\frac{4 + x}{8 + x} = 0.8$$

Step 3: Solve simple way

$$4 + x = 0.8(8 + x)$$

$$4 + x = 6.4 + 0.8x$$

Now solve:

$$x - 0.8x = 6.4 - 4$$

$$0.2x = 2.4$$

$$x = 12$$

✅ Final Answer:

You need to add **12 pints of yellow paint**.

Quick Logic Check:

New yellow = 4 + 12 = 16

New total = 8 + 12 = 20

$$16/20 = 0.8 = 80\%$$

✔ Correct

Agar chaho to main iska ek super-fast shortcut bhi bata sakta hoon jo 10 second me answer deta hai.

why we divide 4+x and 8+x and why we make it equal 0.8 while the above example bananas and apple you just compute the 80% of bananas while here you didnot compute 80& of yello pints nad just simple add it

Excellent 👏 yehi asli confusion hoti hai.

Difference samajh lo — dono questions SAME type ke nahi thay.

🍌 Bananas wala question

Wahan condition thi:

“Bananas 80% more ho jayein”

Matlab:

New bananas = Old bananas + 80% of old bananas

Yani:

$$New = Old \times 1.8$$

Wahan percentage **sirf bananas par apply ho raha tha**.
Total fruits change karna important nahi tha.

Is liye simple 80% nikaal kar add kar diya.

🎨 Paint wala question

Yahan condition kya hai?

“Final mixture ka 80% yellow ho”

Yeh bohat important line hai.

Iska matlab:

Yellow / Total = 0.8

Yeh percentage **total mixture ka part** hai.

😲 Key Difference

Banana question:

> 80% more bananas

Paint question:

> 80% of TOTAL mixture must be yellow

Dono completely different meanings hain.

🔎 Why 4+x and 8+x divide kiya?

Abhi:

Yellow = 4
Blue = 4
Total = 8

Agar x yellow add karte hain:

Yellow = 4 + x
Total = 8 + x

Ab condition lagani hai:

Yellow should be 80% of total.

Mathematically:

$$\frac{\text{Yellow}}{\text{Total}} = 0.8$$

$$\frac{4+x}{8+x} = 0.8$$

Is liye divide kiya.

🧠 Simple Analogy

Socho class me 10 students hain.

Aap chahte ho 80% boys ho.

To kya karoge?

Boys / Total = 0.8

Yahi same logic yahan use hua.

⚡ Sabse Simple Rule

Agar sentence me likha ho:

- “80% more” → multiply 1.8
- “80% of total” → part / total = 0.8

Tumhari thinking bilkul sahi direction me ja rahi hai 👌
Agar chaho to main dono types ka 1 aur example dekar permanently clear kar doon.



↳ New\=Old×1.8

how this 1.8 comes and how this multiplication comes

Very good 👏 ab asli concept clear karte hain.

Tum pooch rahe ho:

New = Old × 1.8 kahan se aaya?

Chalo step by step.

🔷 Step 1: “80% more” ka matlab kya hota hai?

“80% more” ka matlab hota hai:

Old amount + 80% of old amount

Mathematically:

$$New = Old + 80\% \text{ of Old}$$

🔷 Step 2: 80% ko number me convert karo

80% = 80/100 = 0.8

So:

$$New = Old + 0.8 \times Old$$

🔷 Step 3: Old common nikaal lo

$$New = Old(1 + 0.8)$$

$$New = Old(1.8)$$

🔥 Isliye:

$$New = Old \times 1.8$$

🔷 Simple Number Example

Agar bananas = 10

80% more ka matlab:

10 + (0.8 × 10)

= 10 + 8
= 18

Aur agar multiply karo:

10 × 1.8 = 18

Same answer.

ChatGPT 5.2

🔷 General Rule (Yaad rakhna)

Agar:

- 20% more → multiply 1.2
- 50% more → multiply 1.5
- 100% more → multiply 2
- 80% more → multiply 1.8

Formula:

$$\text{More than old} = 1 + \text{percentage in decimal}$$

🔷 Why multiplication use hoti hai?

Kyuki:

Percentage ka matlab hota hai “per hundred”

Aur jab hum “old + percent of old” likhte hain
to algebra me wo multiplication ban jata hai.

Agar chaho to main tumhe “30% less” wala bhi same method se clear kar doon, taake concept complete ho jaye 👌

